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CNSCC.251: Analogue and Digital Communication Systems

Prof. Youping ZHAO
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Lecture 01 Introduction

Agenda

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- **Know each other (Say You, Say Me)**
- **Course Overview (aka. Principles of Communication Systems)**
  - **WHY**: fundamental course for your career + a long list of applications (just to name a few)
  - **WHAT**: block diagram of typical communication systems + knowledge graph
  - **HOW**: critical thinking + active learning
  - **Syllabus**: course outline + class/lab schedule
  - **Textbook:** reference books and online MOOC resources
  - **Assessment:** quiz, lab reports, final exam
  - **Call for raising critical questions, active involvement into the class, initiative/volunteer contributions….**
- **Homework**

Say You, Say Me

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- **My brief bio.**
  - Educational
    - BSEE – MSEE – Ph. D.
  - Working experience
    - 1G-2G-3G-4G-5G
    - Cognitive radio-AI-6G
  - Editor for JCIN， China Communications
  - Books, papers, patents
  - **Ph.D. & Master graduate student advisor**
  - Enjoy traveling, jogging

- **How about each of you?**
- Brief introduction of yourself in English
  - hometown
  - **Research interests** or favorite course
  - hobby
  - goal after graduation, if you'd like to share
  - **Do you like thinking & asking critical questions in the past classes**?

Instructor： Prof. Youping ZHAO

School of Electronics and Information Engineering

Beijing Jiaotong University

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- Webpage: http://faculty.bjtu.edu.cn/8376/

Brief Introduction: Prof. Zhao Youping

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清華大學 Tsinghua University

1994 BSEE
1996 MSEE

VirginiaTech Invent the Future

2007 Ph.D.in EE



北京交通大學 Beijing Jiao Tong University

2010-now Professor

Research Interests: **over 10 years overseas study/work experience in USA**

5G/6G wireless communications, channel characterization, cognitive radio and intelligent communication systems

Historic View of Wireless Communications

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"To understand a science it is necessary to know its history"

—Auguste Comte (1798–1857)

- **1G: Analog Radio** ~ 4 kHz (mainly voice)
- **2G: Digital Radio** (narrowband) ~ 10-100 kbps
- 3G: Digital Radio (wide band) ~ 2 Mbps
- 4G: **Software Defined Radio** (broadband) ~ 100 Mbps
- **5G~6G: Intelligent/Cognitive Radio** > 1 Gbps



Course Overview

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Outline Syllabus

This module has been designed , mainly from a systems point of view , as a comprehensive introduction to analogue and digital communication systems in general and to physical layer in particular. The following topics are therefore covered:

Introduction to Communication Systems and Data Networks; Concepts, models and systems. Signals: analogue, digital signals, bandwidth and bit rate.

Noise types and evaluation. Aspects of Probability and Statistics. Circuit noise, noise measures, Noise factor, Noise temperature and its measurement. Cascaded systems.

Advanced mathematical calculus such as Fourier analysis.

Analogue and digital transmission, transmission impairments and channel noise. Shannon communication capacity.

Analogue Transmission, modulation and demodulation of analogue signals, AM, FM and PM techniques.

Digital Transmission, Aspects of Source Coding, Line Coding and Channel Coding

Modulation of digital data, Principles of ASK, FSK, PSK QAM, Introduction to Modems.

Aspects of Multiplexing techniques, FDM, DTM, CDM and Multiple Access Schemes

Let us learn this course altogether and have fun!

Aims of this Course

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Educational Aims of the Module

i. Subject Specific

Introduce students to Analogue and Digital Communications at a system level and enable students to understand the role that each components plays in maximizing system performance;
Enable students to understand the stochastic nature of noise in general and its effect on both the design and performance of analogue and digital communication systems;
Present students with aspects of data communication networks in terms of overall design and usage;
Introduce students the main concepts of source and channel coding;
Familiarize students with analogue and digital modulation schemes and aspects of design and implementation;
Familiarize students with multiplexing and multiple accessing techniques;
Provide students with a broad range of knowledge and skills necessary for a career in communication engineering.

ii. General

Expose students to problem-solving and investigative techniques;
Help students to critically i) evaluate technical ideas and ii) compare system designs and performance;
Assist them in communicating their conclusions to both expert and non-expert audiences;
Enhance their independent study as well as their presentation and interpersonal skills.

Expected outcomes of this course

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Intended Learning Outcomes of the Module

i. Subject Specific

At the end of the course the strudents should be able to:
Demonstrate their knowledge of concepts as well as of mathematical formulations of communication methods, related systems and components, at both the transmitting and receiving ends.
Identify, analyse and model noise processes, which introduce errors to transmitted signals
Develop software that implements the described communication methods.

ii. General

On successful completion of this module students will be able to:

Manage their study time efficiently and demonstrate independent learning abilities required for continued professional development.
Acquire and demonstrate analytical skills
Successfully integrate diverse information to form a coherent understanding of the subject.
Critically reflect on technical advancements. Demonstrate effective written and oral skills.

Assessment

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Assessment Proportions

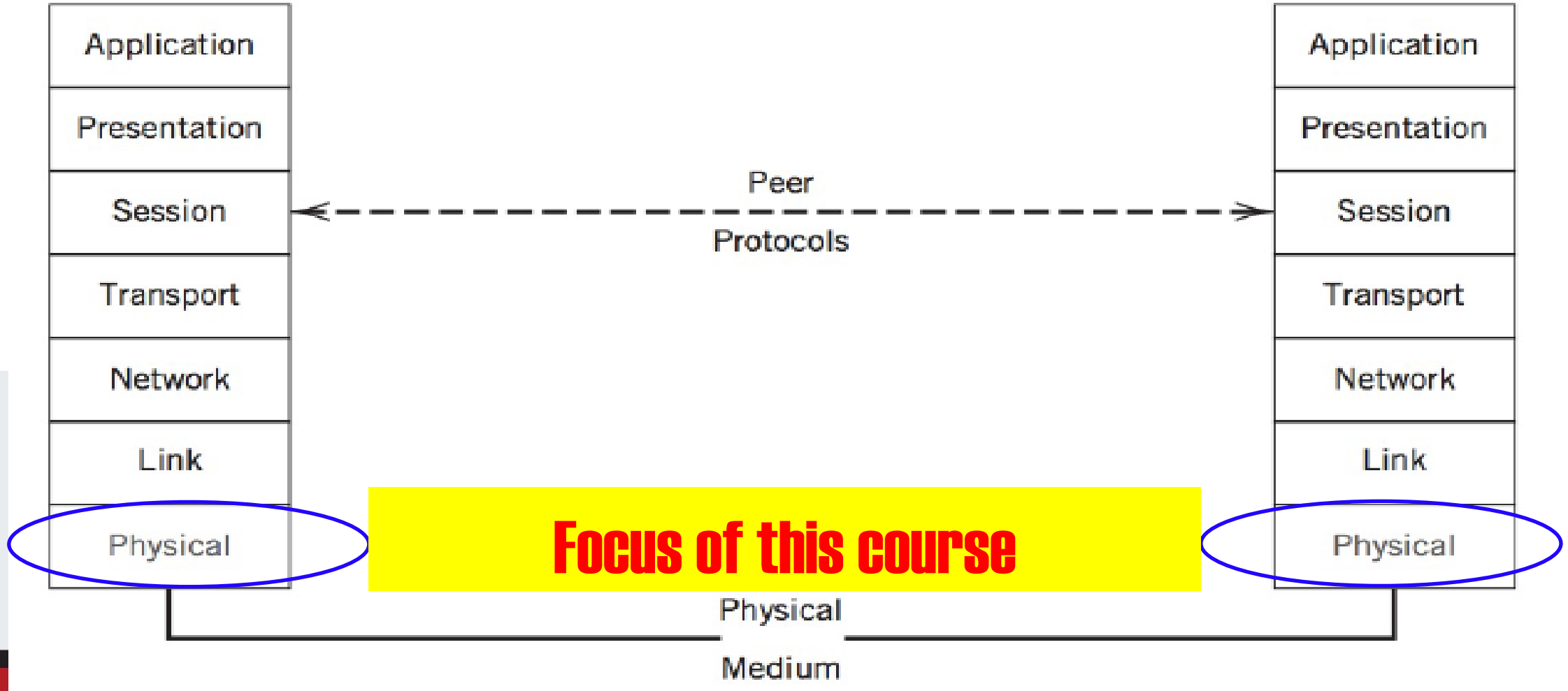
Assessment Type	Status	Percent
Exam	Compulsory	60
Coursework	Compulsory	40

Coursework: 6 Labs + Lab reports

Let us learn this course altogether and have fun!

OSI Model of Communication Systems

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OSI: Open System Interconnection…. ISO model…

Source: S. Haykin, ***Communication Systems***, 5th Edition, Wiley, 2008

Course Outline

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1. Introduction to Communications Systems
   - ✓ Signal & Noise Analysis (Random Process)
   - ✓ Basics of Information theory (especially, Shannon information capacity)
2. Analog (Continuous-Wave) Modulation
   - ✓ AM, FM, PM;
   - ✓ Noise effect in analog modulation
3. Digital (Pulse) Modulation: PCM
4. Digital Baseband/Passband Transmission: ASK, PSK, FSK, QAM
5. Channel Coding, Multiplexing, and Multiple Access Schemes

Objectives

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- **Understand** how a communication system works via modelling
- **Analyze** the performance of a communication system
- **Understand** the pros and cons of various communication systems and **learn** how to design the appropriate communication system for a given application

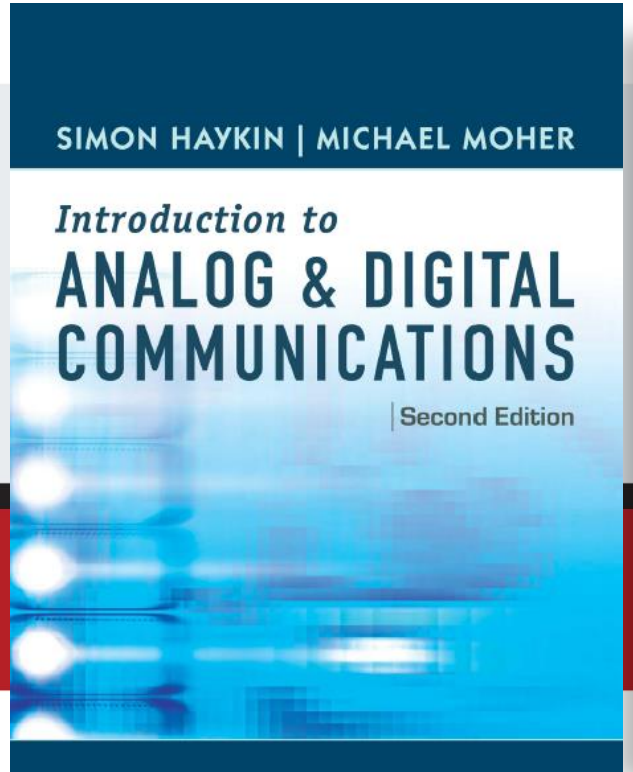
- **Prerequisites**
  - ✓Signals and Systems
  - ✓Probability and Statistics
  - ✓Calculus, Linear algebra

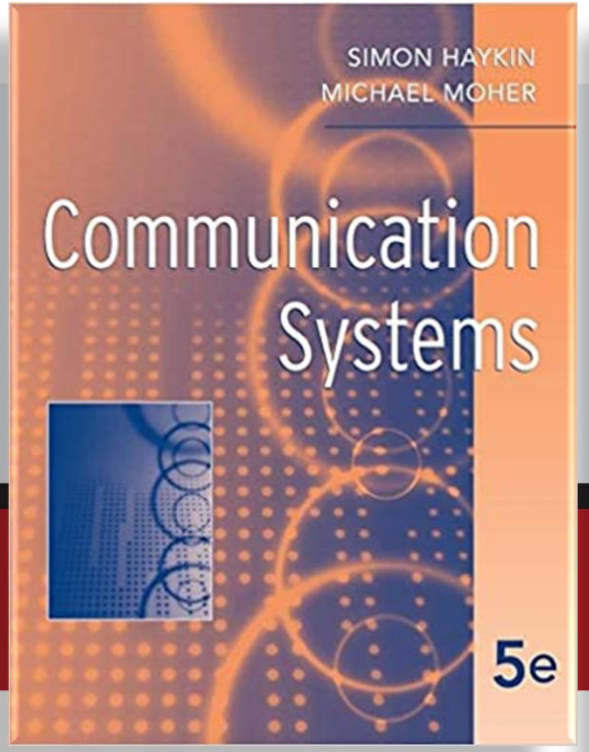
Play 5G MIMO video

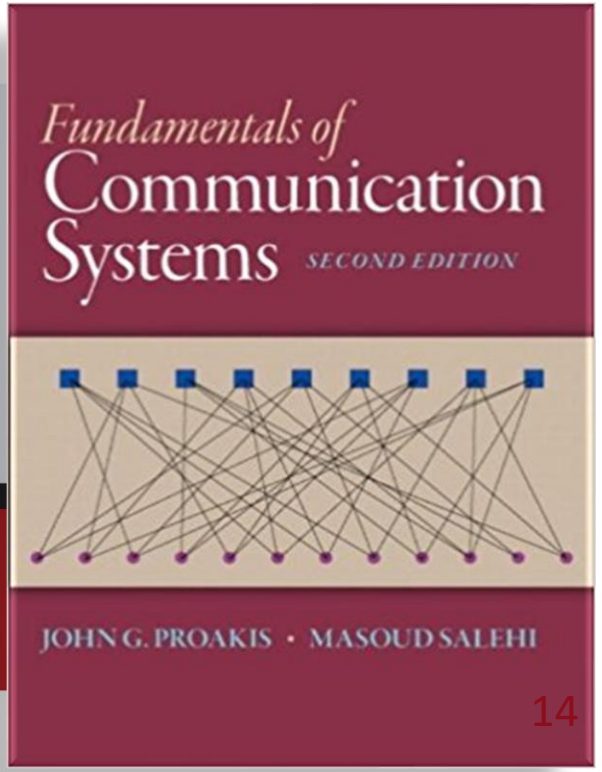
Textbooks and References

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- **Introduction to Analog and Digital Communications,** by Simon Haykin
- **Communication Systems**, 5th Edition, by Simon Haykin, John Wiley & Sons, Inc.
- **Fundamentals of Communication Systems**, 2nd Edition, by John Proakis, Masoud Salehi, Pearson



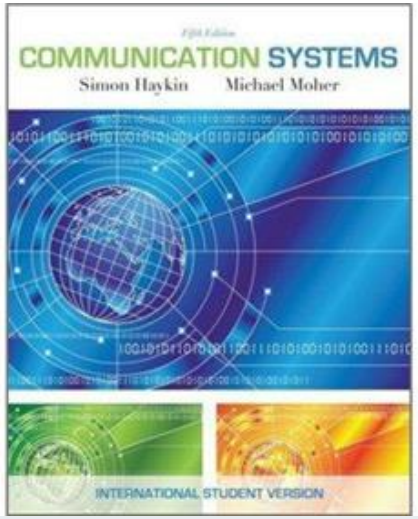




Alternative Textbooks

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Behrouz A Forouzan , Data Communications and Networking, MacGraw Hill , January 2009
Bernard Sklar, Digital Communications: Fundamentals and Applications, Prentice Hall, 2001



Communication Systems, 5th Edition

S. Haykin, John Wiley & Sons, Inc., New York, 2010



Communication Systems, 4th edition

S. Haykin, John Wiley & Sons, Inc., New York, 2001

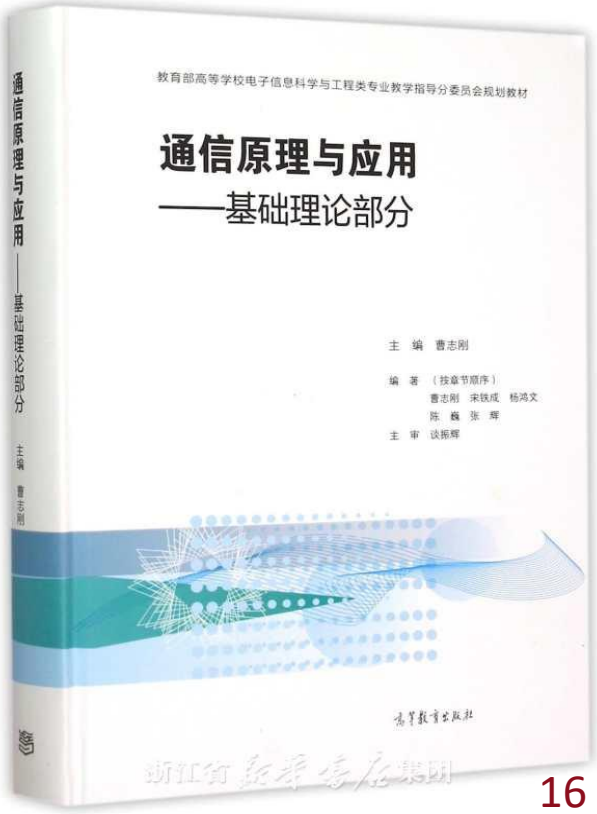
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Reference books

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Other reference books (in Chinese) :

- 郭宇春，通信系统原理，科学出版社
- 曹志刚，通信原理与应用-基础理论部分，高等教育出版社
- 周炯槃，通信原理（第3版），北京邮电大学出版社
- 樊昌信，通信原理（第6版），国防工业出版社



Cross-reading is highly suggested

Other resource (e.g., MOOC)

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- **Shanghai Jiaotong University**: (Lectures by Prof. Tao Meixia)
  http://iwct.sjtu.edu.cn/personal/mxtao/teaching.html

- **MIT**:
  http://open.163.com/special/opencourse/information.html

- **Crowdsourcing for quality course materials** (part of homework for this week)

Homework

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- **Read** the **Background & Preview** and the **Chapter 1** of **Haykin's textbook**
- **Search** for **online course materials**, PPT, videos, website from Top Universities around the world (e.g., MIT, Stanford, Georgia Tech., SJTU, Tsinghua, etc.) and write a survey with a list of your findings

  [You will talk about your findings in upcoming classes]

  – MOOC (massive open online courses)

Learning Groups

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- **2 Class Representatives:**
  - Dragon-1 (Class 2101): Weiyang Jin
  - Dragon-2 (Class 2102): Xinyi Huang
- **8 Learning Groups: 8 group heads are TBD**

Problem-Driven Learning

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- **Questions in your mind before taking this course?**
  - Each student talks about his/her question(s) related to “communication systems”
- **This course aims to answer the following 3 questions?**
  - What is communication?
  - What does a communication system consist of?
  - How to design a good communication system?

Fundamental Questions

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- Is there a general methodology for designing communication systems?





- Is there a limit to how fast one can communicate?



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Communicate:

To make known;

Convey, impart, spread, transfer, transfuse,
Broadcast, diffuse, disseminate, propagate,
Contaminate, infect, poison

Ref: Webster Dictionary

What is communication?

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"The systems and processes that are used to convey information from a source to a destination **efficiently and reliably**, especially by means of electricity or radio waves."



Also including light wave

Source: S. Haykin, ***Communication Systems***, 4th Edition, Wiley, 2001

What is communication?

- technical definition
- Tele-communication （Tele： far） is the exchange of signs, signals, messages, words, writings, images and sounds or information of any nature by wire, radio, optical or other electromagnetic systems. ——ITU

  - Transmit messages with electrical signals in spatial-temporal distance

- "Article 1.3" (PDF), ITU Radio Regulations, International Telecommunication Union, 2012
- Constitution and Convention of the International Telecommunication Union, Annex (Geneva, 1992)

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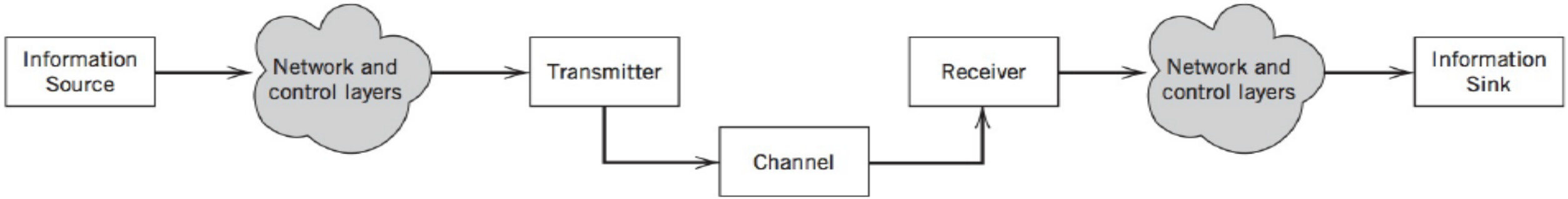
Communication Process

In the most fundamental sense, communication involves implicitly the transmission of *information* from one point to another through a succession of processes, as described here:

1. The generation of a *message signal*: voice, music, picture, or computer data.
2. The description of that message signal with a certain measure of precision, by a set of *symbols*: electrical, aural, or visual.
3. The *encoding* of these symbols in a form that is suitable for transmission over a physical medium of interest.
4. The *transmission* of the encoded symbols to the desired destination.
5. The *decoding* and *reproduction* of the original symbols.
6. The *re-creation* of the original message signal, with a definable degradation in quality; the degradation is caused by imperfections in the system.

Source: S. Haykin, ***Communication Systems***, 4th Edition, Wiley, 2001

Elements of a communication system

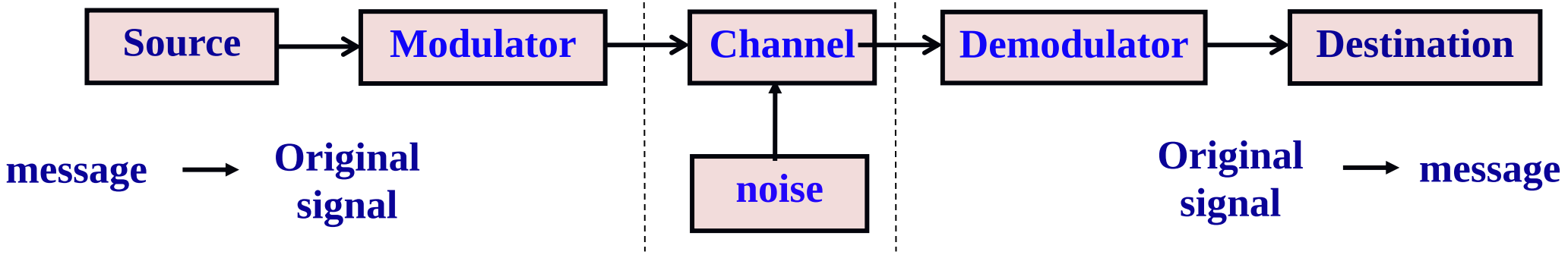


Source:

S. Haykin, ***Communication Systems***, 5th Edition, Wiley, 2008

Analog Communication System Model

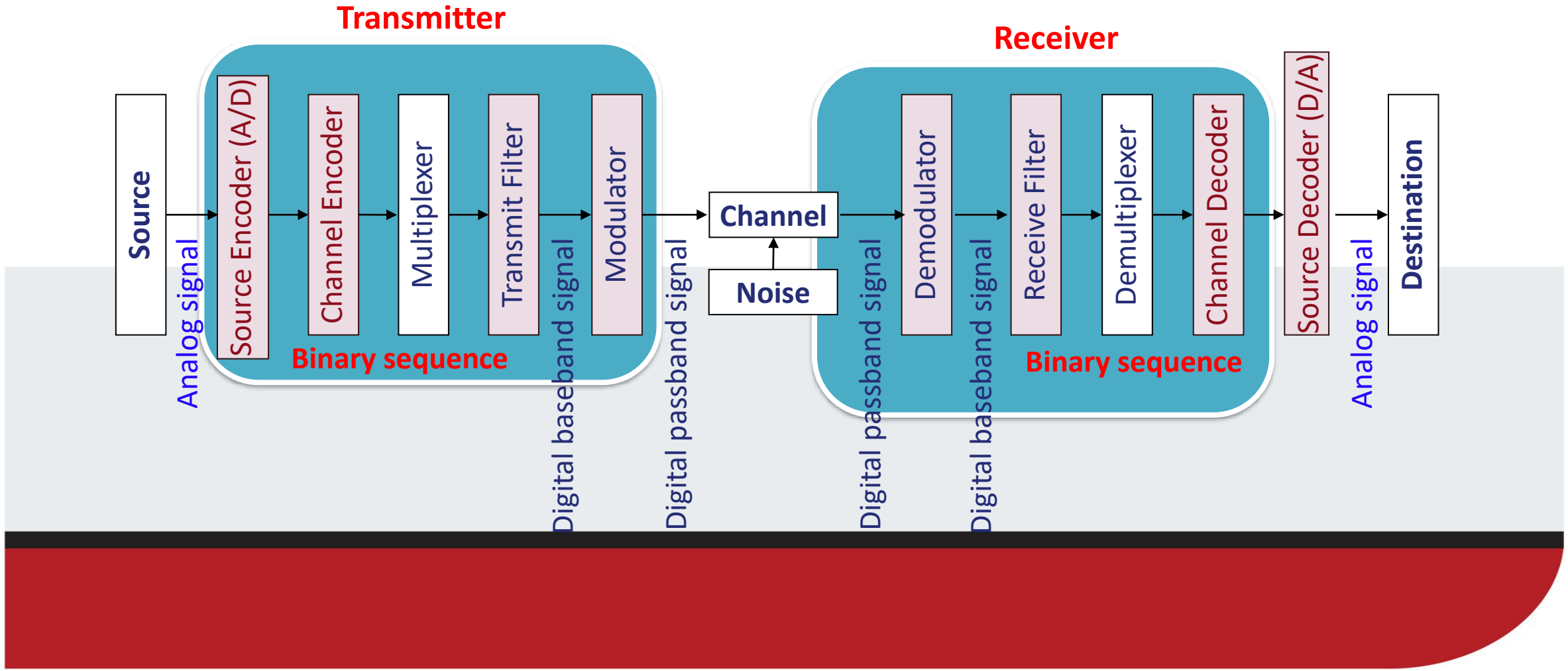
- Transmitting analog signal



  - Modulation
    - Changing one or more features for a carrier (periodic signal) according to the modulating signal (to be transmitted with information)
  - Demodulation
    - Extracting the original signal with information from the modulated carrier

Digital Communication System Model

- Transmitting digital signal



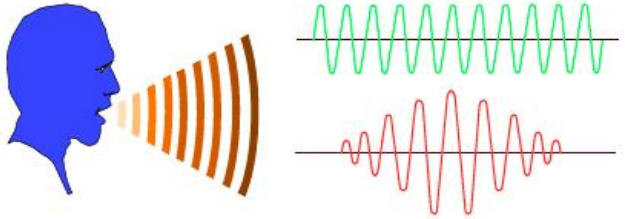
Types of Communication Systems

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- Analog Communications







- Digital Communications



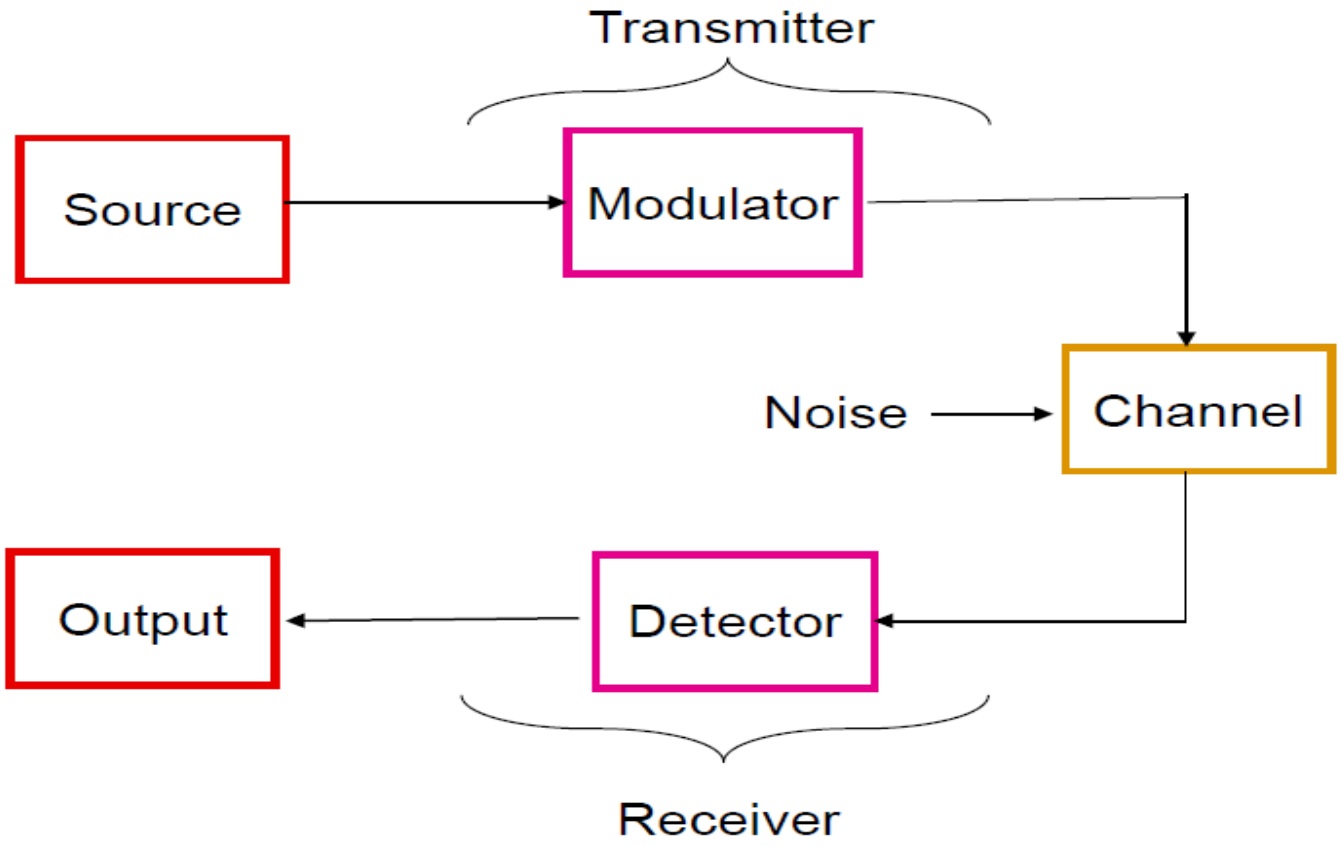
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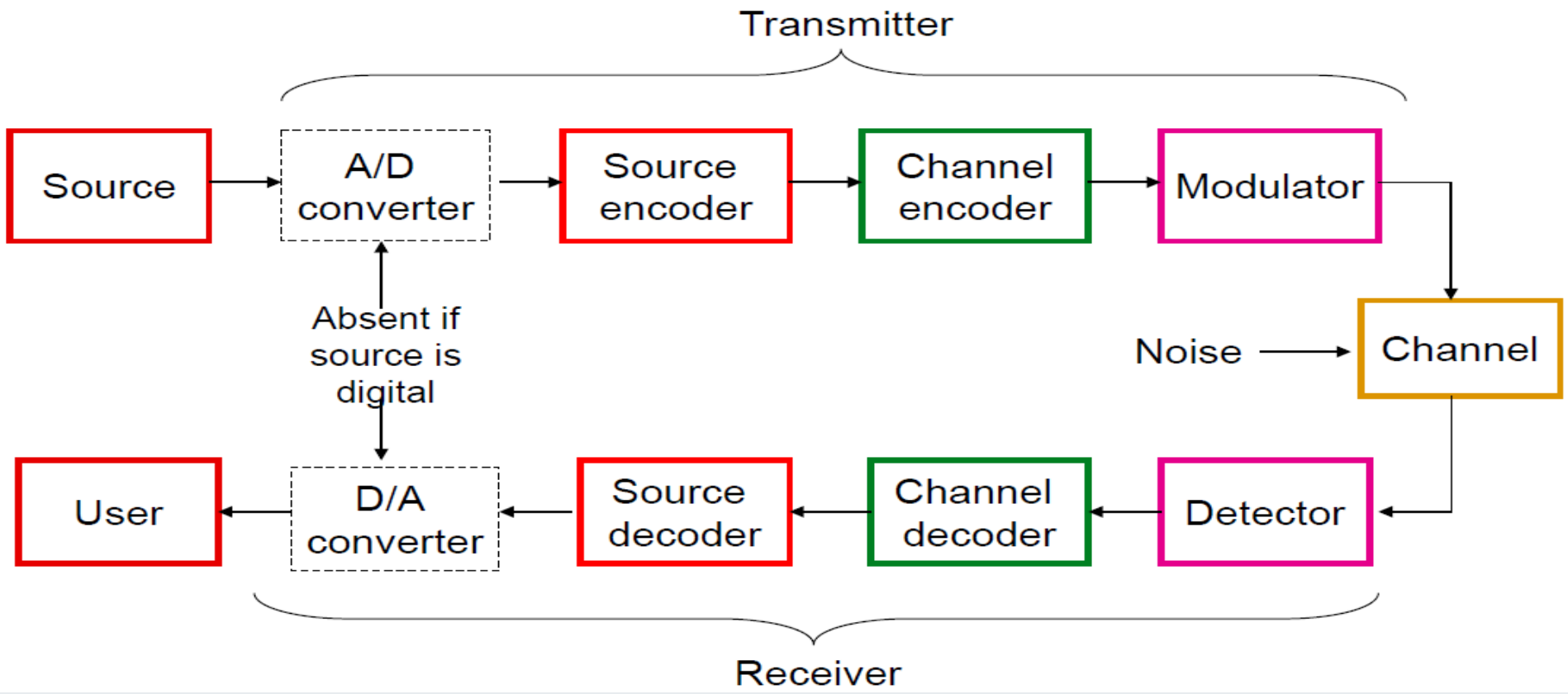




Analogue Communication System Model



Digital Communication System Model



Why Digital Communication System?

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- Robustness to channel noise and external Interference



- Security of information during its transmission from source to destinatio



- Integration of diverse sources information into a common format



- Low cost DSP chips by very cheap VLSI designs



Faster
Higher
Stronger
Altogether

Measure of Commu. System Performance?

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- **Information measure**
  - The amount of information to be transmitted in a communication system
- **Efficiency measure**
  - The efficiency of maximizing the data transmission rate given transmission bandwidth
- **Reliability measure**
  - The correctness of information transmission given signal power in noisy channel

Performance Metrics of Communication Systems

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- Reliability
  - SNR for analog systems
  - Bit error rate for digital systems
- Efficiency
  - Bandwidth efficiency
  - Energy efficiency



Performance Tradeoff

Digital Communication System Model

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Bandwidth Efficiency

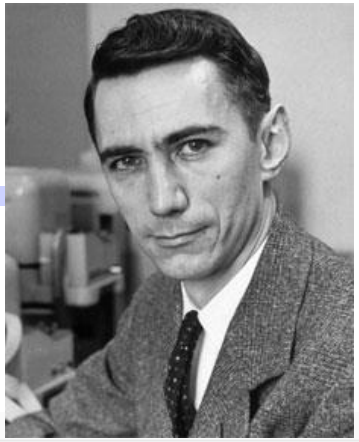
$$\text{bandwidth efficiency} = \frac{\text{data rate } R}{\text{bandwidth } W} \text{ bits/sec/Hz}$$

Play
5G MIMO video
again

Energy Efficiency

$$\text{energy efficiency} = \frac{\text{bit energy}}{\text{noise power spectral density}} = E_b/N_o$$

Essence of communication?





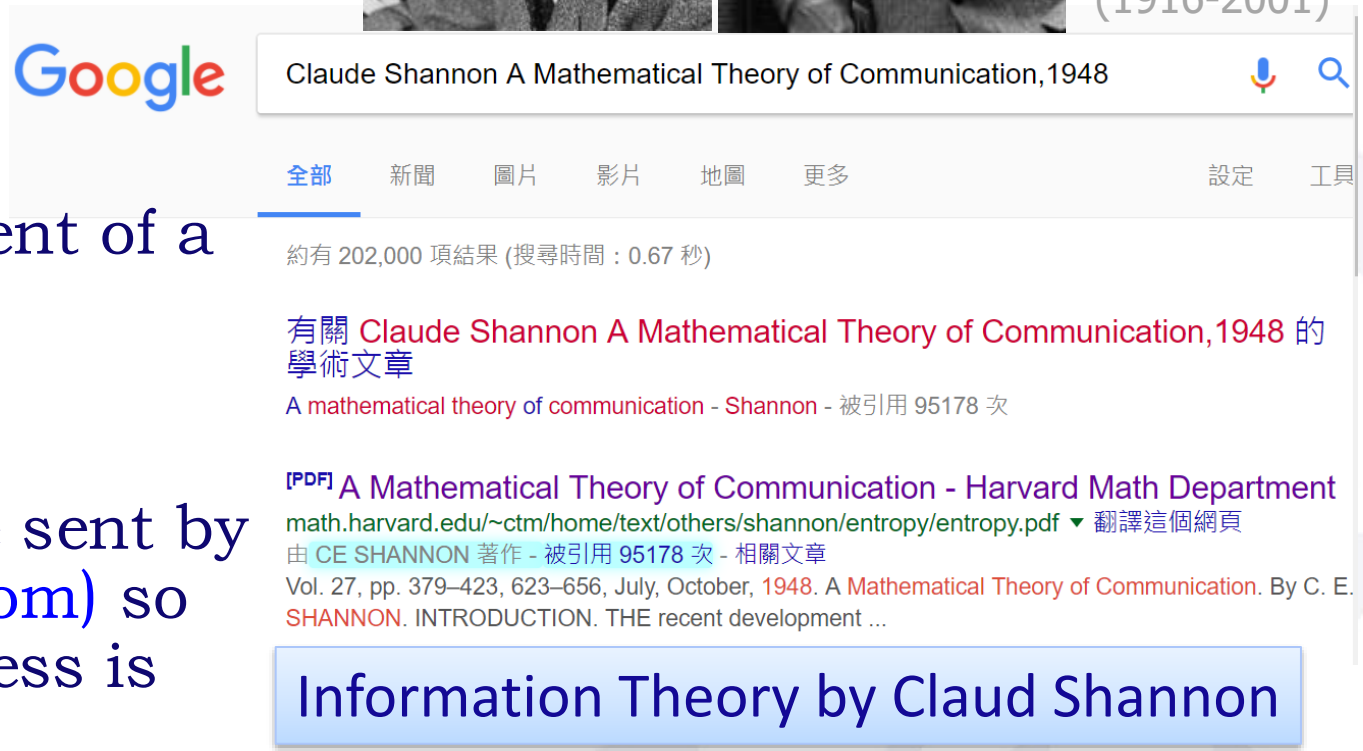
(1916-2001)

- Is the message meaningful to the recipient?

  Msg 1: It is daytime now.

  Msg 2: Our classroom is 2-C107.

- **Information(信息)** is the content of a message that can **eliminate uncertainty** for the receiver
- The stochastic nature of communication: The message sent by the source is uncertain (random) so that the communication process is meaningful to the receiver



Information Theory by Claud Shannon

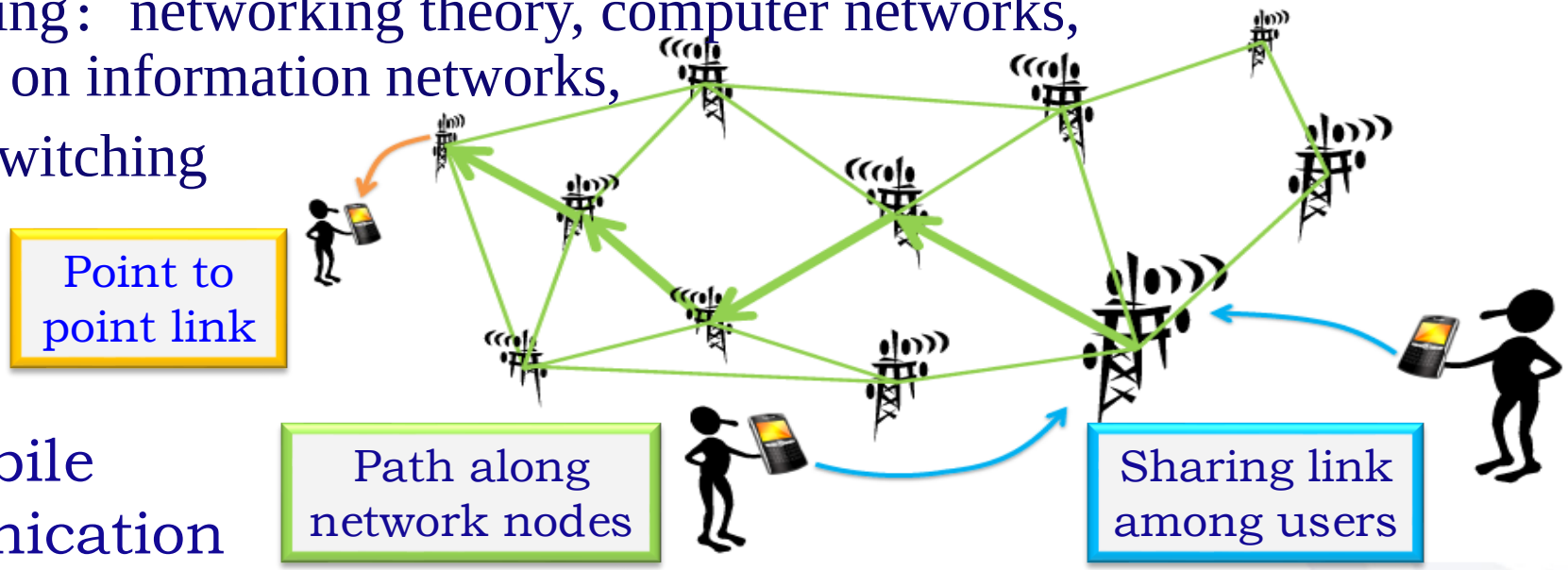
Theoretical basis of modern communication

This course versus others

- Communication systems: information transmission principles and implementation methods
- Point-to-Point communication systems
- Channel sharing： digital communication, wireless communication
- Networking： networking theory, computer networks, seminars on information networks, modern switching

Communication systems

Communication networks



Mobile communication

COMMUNICATION AND INFORMATION

- **Information** is the content of the message that can eliminate uncertainty for the receiver
- **Communication** is the process of transmitting information using electrical signals, which is inherently random
- This course is focused on the **principles** of **point-to-point communication** systems

1.2 MODEL OF COMMUNICATION SYSTEM

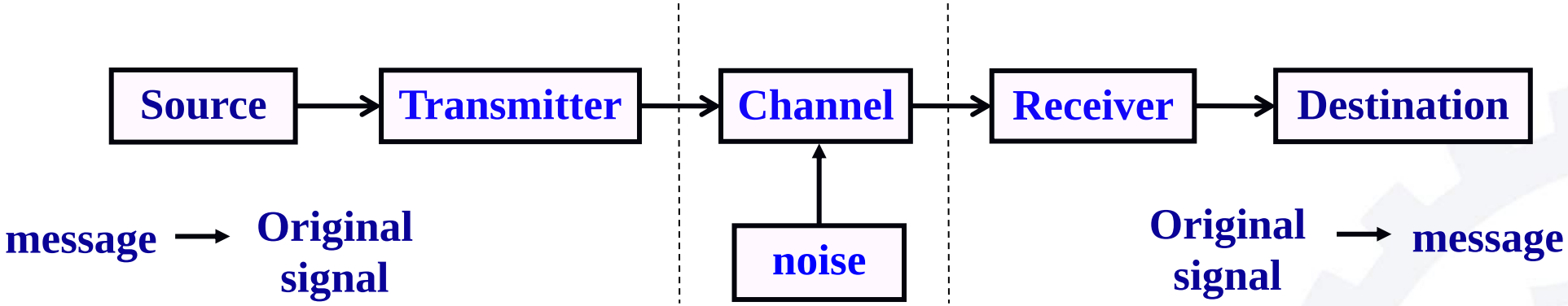
A model of general communication system
A model of analog communication system
A model of digital communication system

Communication system

- Communication system consists of
  - ✓transmitting/receiving equipment and transmission media (channel) to complete the communication process
- How to model a communication system?
  - ✓extract key factors, describe the relationship between various factors in mathematical language, and obtain meaningful results
- Why need to study the modeling?
  - ✓Describe the principles of communication systems and analyze the performance of the communication systems
  - ✓Explore theoretical limits and guide engineering practice

Model of communication system

- A general model



What transmitters and receivers should be used to send different types of signals?

Digital signal vs. analog signal

Digital signal

Limited messages in a period of time

Temporal discrete signal

Limited signal： 0/1

Analog signal

Unlimited value of messages

Temporal continuous signal

Messages： real value

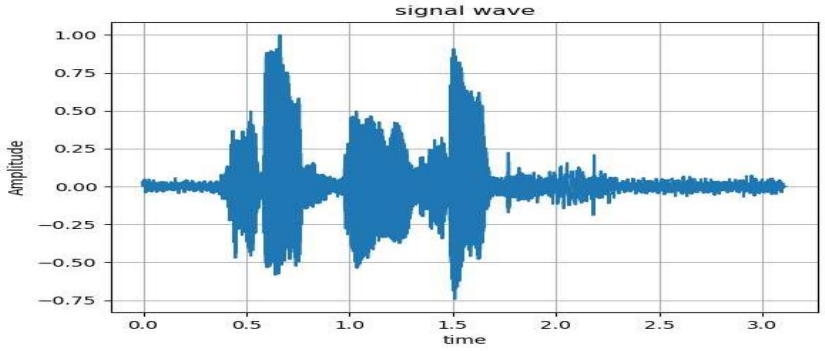
- Discrete symbol sequence, e.g. text message

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S O S

Telegraph: The earliest digital communication method that uses electrical signals to transmit text messages

- Continuous wave, e.g. voice message

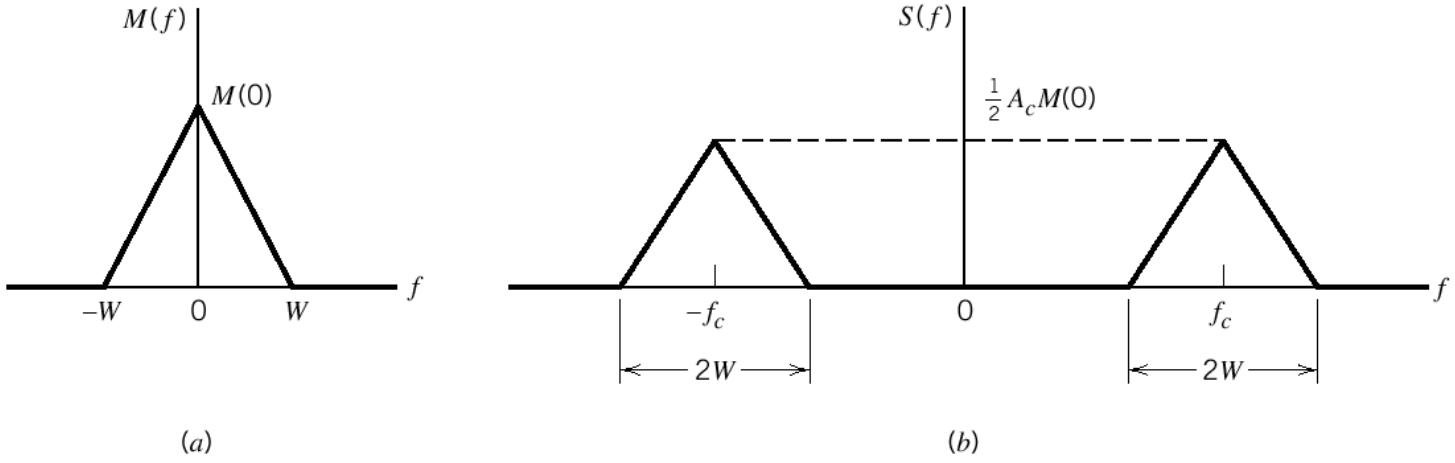


Telephone: The earliest analog communication method using electrical signals to transmit voice messages

Baseband transmission vs. Passband transmission

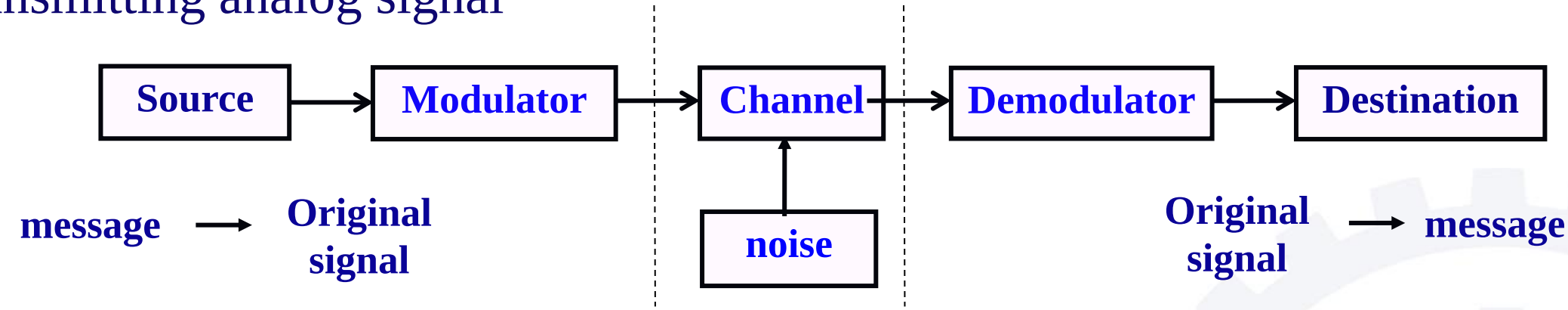
- Baseband signal
  - Signals with a spectrum around zero frequency
- Baseband transmission
  - Transmitting baseband signal

- Passband signal
  - Signals with a spectrum around high frequencies
- Passband transmission
  - Transmitting passband signal which is modulated signal



Analog Comm. Sys. Model

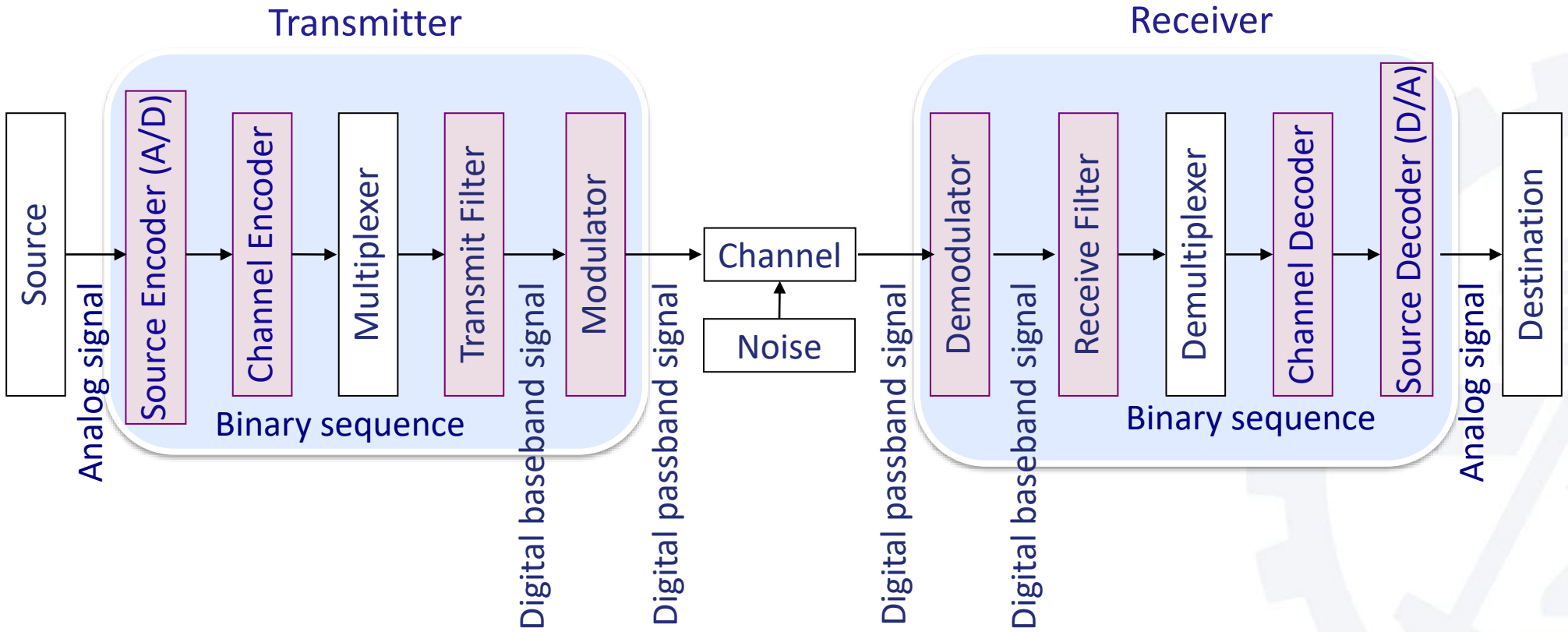
- Transmitting analog signal



  - Modulation
    - Changing one or more features for a carrier (periodic signal) according to the modulating signal (to be transmitted with information)
  - Demodulation
    - Extracting the original signal with information from the modulated carrier

Digital Comm. Sys. Model

- Transmitting digital signal



Types of Communication System

- Signal
  - Analog/Digital
- Transmission
  - Baseband/Passband
- Media (Channel)
  - Wire/Wireless
- Direction
  - Simplex (radio, television)
  - Duplex (telephone)
  - Half-duplex (two-way radio)
- Service
  - Telephone/telegram/data/video…
- Parties
  - 2 /*N* users
    - point to point communication system
    - Any pair of users in a network
    - 1 transmitter, many receivers in broadcast system



Exercise:
Is there any other perspectives for classifying communication systems?

COMMUNICATION SYSTEM MODEL

Key factors and their relationship in a communication system

Models of communication systems: general/analog/digital

- Information Measure
- Efficiency Measure
- Reliability Measure

1.3 PERFORMANCE OF COMMUNICATION SYSTEMS

Measure of the performance?

- Information measure
  - The amount of information to be transmitted in a communication system
- Efficiency measure
  - The efficiency of maximizing the data transmission rate given transmission bandwidth
- Reliability measure
  - The correctness of information transmission given signal power in noisy channel

1.3.1 INFORMATION

Information

- Given a set of two messages, which message is of more information?

  Message 1： The classroom is at 2-C107.

  Message 2： Quick! Got a Test!

- How to create a metric, $I(x)$, to measure the information in a message?
  - The randomness of information

    A function of $P(x)$, the probability of message x
  - The more extent of uncertainty is removed when a less possible message is received

    A reversal function of $P(x)$
  - The information of multiple independent messages should be cumulative

    A logarithm function is needed to make joint probability of messages additive

$$I(x) = -\log_a P(x)$$

Unit of information

- Information of message x

$$I(x) = -\log_a P(x)$$

- Unit
  - Information is unitless, an indicative unit is used
  - bit（比特）， $a=2$
  - nat（奈特）， $a=e$
  - hatley（哈特莱），$a=10$

- e.g. Binary 0/1 massages, P(0)=P(1)=1/2. The information of message 1 and message 0?

Solution:

$$I(x) = -\log_2 1/2 = 1\text{bit}$$

Message 1 and message 0 are both of 1 bit information

Average Information

- Given a set of *M* symbols $x_i,\ i = 1, \cdots, M$
  the average information of each symbol is

$$H(X) = -\sum_{i=1}^{M} P(x_i) \log_2 P(x_i)$$

  - For the uniform distribution of messages.

$$H(X) = -M\left(\frac{1}{M}\log_2\frac{1}{M}\right) = \log_2 M = k \text{ bit} \quad (M = 2^k)$$

- also called entropy（熵, 热力学概念，分子运动的混乱程度）
  - The larger the entropy, the more confused (in thermodynamics), the larger the uncertainty (information theory)
  - The entropy of a Chinese character is 9.65 bit, but that of a English letter is 4.03 (removed the redundancy)
    - Indicating that Chinese is more complicated than English

Example

Given a source of information with 4 symbols of the probability $\begin{pmatrix} A & B & C & D \\ 1/2 & 1/4 & 1/8 & 1/8 \end{pmatrix}$

Comparing with another information source of 4 equal likely symbols, which one is of larger entropy?

Solution: The entropy of this information source

$$H(X) = -\sum_{i=1}^{4} P(x_i)\log_2 P(x_i)$$

$$= 1/2\,(\log_2 2) + 1/4\,(\log_2 4) + 2 \times 1/8\,(\log_2 8)$$

$$=1/2+1/4 \times 2+ 2 \times 1/8 \times 3 \ = 7/4 \text{ bits}$$

The entropy of corresponding equal likely source set

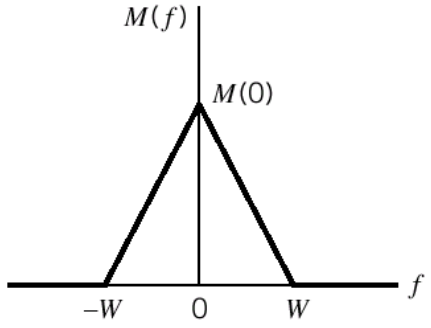
$$H(X) = \log_2 M = \log_2 4 = \ 2 \text{ bits}$$

The latter information source has higher entropy.

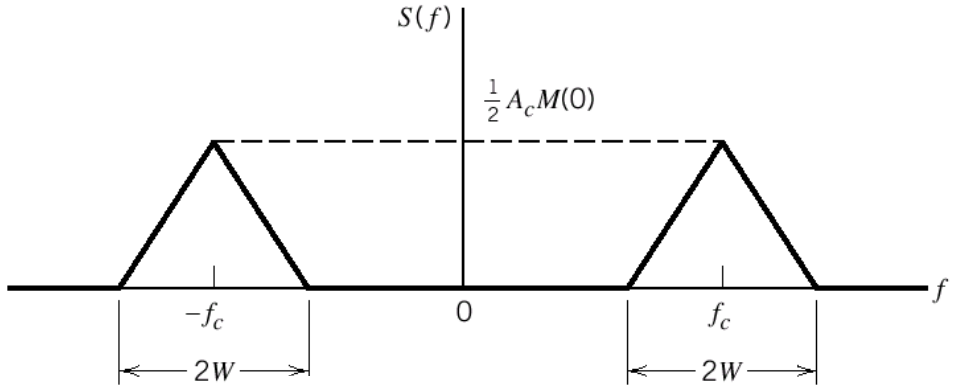
1.3.2 EFFICIENCY

Efficiency

- For an analog system
  - Effective bandwidth, B: the frequency range covering the most (90%) harmonic frequencies
    - unit：Hz 赫茲



Bandwidth
B = *W*



Bandwidth
B =2 *W*

The measure of efficiency

- For a digital system
  - Information rate 传信率， R_b
    - Also called bit rate 比特率
    - Unit: bps, bits/s, 比特/秒
  - Symbol rate传码率，符号速率，R_B
    - Unit: Baud, Bd, 波特
    - Symbol ~ 消息

$$R_b = R_B H$$

$$H = \log_2 M$$

$$R_b = R_B \log_2 M$$

H: average information of a M-ary symbol for the equal likely case

Speech (GSM): 9.6 kbps, 4K video peak rate: 10 Mbps
5G mobile phone download rate: 1.4 Gbps

Exercise： What is the information rate of the typical communication services?

Example

Given a binary digital signal, 72000 symbols are sent in 2 minutes, find

(1) the information rate and the symbol rate respectively;

(2) the information rate with a 8-ary digital signal with the same symbol width (in time).

Solution:

(1) R_B=72000/120=600 Baud; R_b=R_B=600 b/s

(2) For the 8-ary signal, the same symbol width means the same symbol rate, hence the information rate is R_b=$R_B$$\log_2$M= 1800 b/s

The measure of efficiency

- For a digital system
  - Information rate， R_b
    - 传信率, bit rate 比特率
    - Unit: bps, bits/s, 比特/秒
  - Symbol rate, R_B
    - 传码率，符号速率
    - Unit: Baud, Bd, 波特
    - Symbol ~ 消息
  - Bandwidth efficiency, η
    - 频带利用率
    - Unit：b/s/Hz

$$R_b = R_B H$$

$$H = \log_2 M$$

$$R_b = R_B \log_2 M$$

H: average information of a *M*-ary symbol for the equal likely case

$$\eta_b = \frac{R_b}{B}$$

e.g.： Which system is more efficient?
A: R_b = 2Mb/s, B=2MHz； B: R_b = 1.5Mb/s, B=1MHz

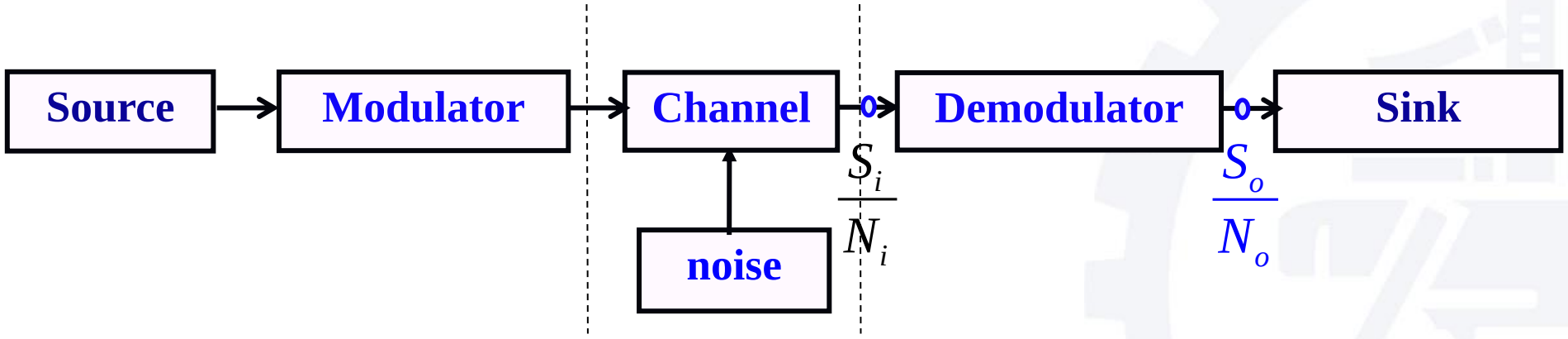
Solution: system A, $\eta_b = \frac{R_b}{B} = 1$ b/s/Hz;

system B, $\eta_b = \frac{R_b}{B} = 1.5$ b/s/Hz

1.3.3 RELIABILITY

Reliability

- For an analog system
- Output Signal-to-Noise power Ratio (SNR), S_o/N_o
  - 输出信噪（功率）比
  - Logarithmic value： $10\lg(S_o/N_o)$, Unit: decibel 分贝 （dB)



Reliability

- For a digital system
  - Symbol error rate 误码率

$$P_e = \lim_{N\to\infty} \frac{n, the\ number\ of\ error\ symbols}{N, the\ number\ of\ tranmitted\ symbols}$$

  - Bit error rate 误比特率

$$P_b = \lim_{N_b\to\infty} \frac{n_b, the\ number\ of\ error\ bits}{N_b, the\ number\ of\ tranmitted\ bits}$$

e.g.: Given an 8-ary digital communication system with bit rate 12000 b/s,
216 error symbols are found in half an hour at the receiver, find the symbol error rate

Solution: R_b = 12000 b/s ; $R_B = R_b/\log_2 8$ = 4000 Baud;
$P_e = 216/(R_B \times 30 \times 60) = 3\times10^{-5}$

Question:
Is it possible to improve the efficiency and reliability of a communication system simultaneously?

PERFORMANCE OF A COMM SYSTEM

Average information: the amount of information

efficiency: effective bandwidth, transmission rate

reliability: signal-to-noise ratio, bit error rate

Channel and interference

Channel model

Channel capacity

1.4 CHANNEL AND INTERFERENCE

1.4.1 CHANNEL AND INTERFERENCE

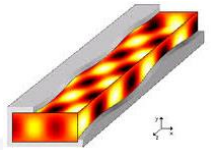
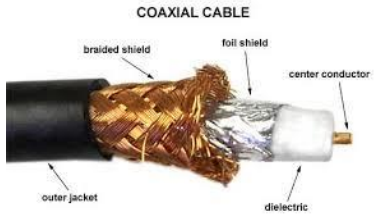
Channel

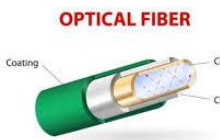
- Transmission media
  - Some physical media for transmitting electrical or optical signal
  - Wire channel
    - wire, twisted pairs, coaxial cables, wave guides: for electrical signals.
    - Optical fiber: for laser signal.
  - Wireless channel
    - Space (atmosphere, vacuum): for electromagnetic signals.
- Storage media
  - Discs, disks, flash drives, and SSDs, etc.

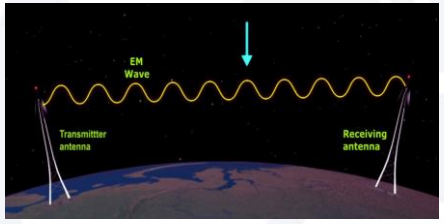













Attenuation and interference

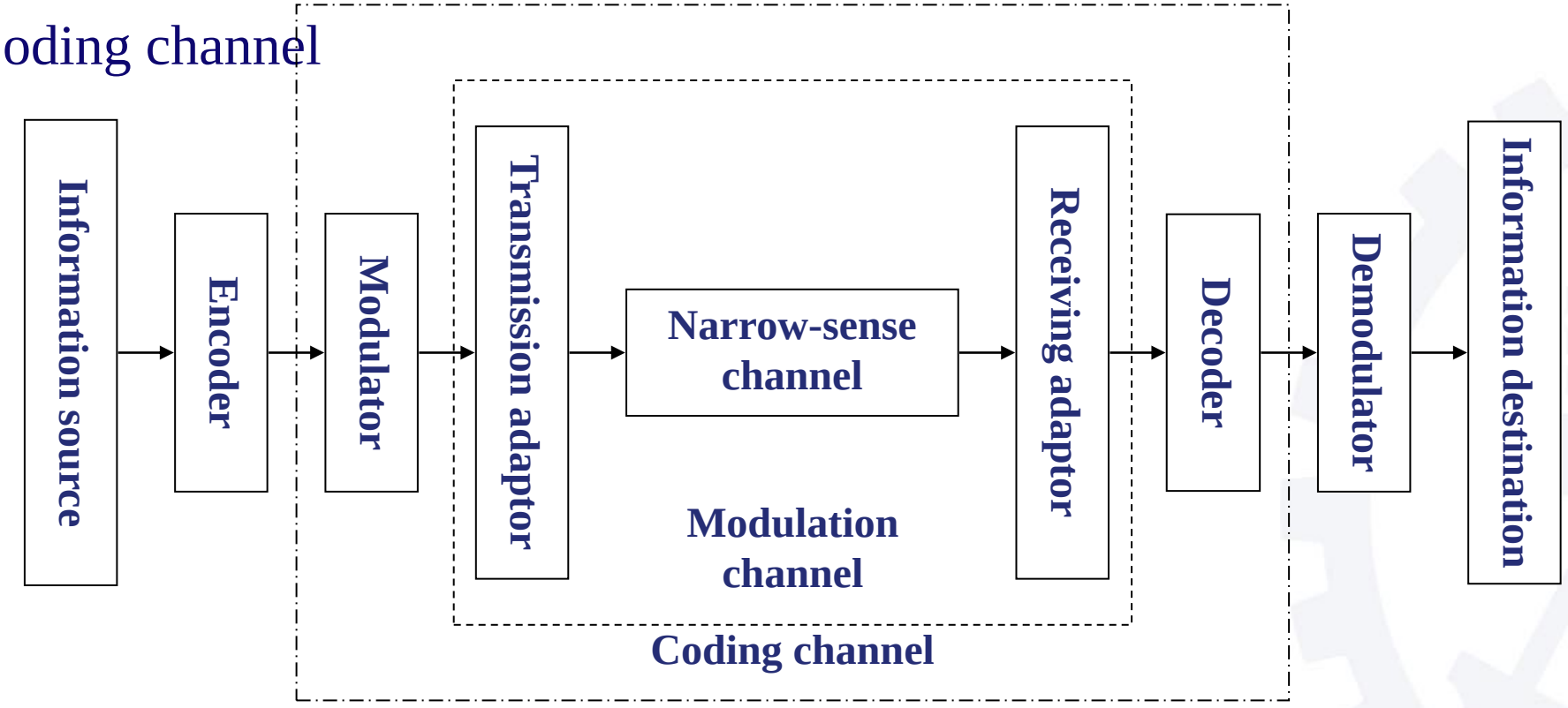
- Signal attenuation： grows with the transmission distance
  - The wire resistance increases with length
  - Radio waves diffuse with distance and are absorbed by blockers
- Noise interference: causes signal distortion
  - Noise: Thermal noise of components inside the communication system
  - Interference: The external signal of the communication system
    - Free space openness, electromagnetic induction.
    - Natural interference: lightening
    - Man-made interference: other signal, electromagnetic interference.

Question: How to describe the effect of a channel on a signal?

1.4.2 CHANNEL MODEL

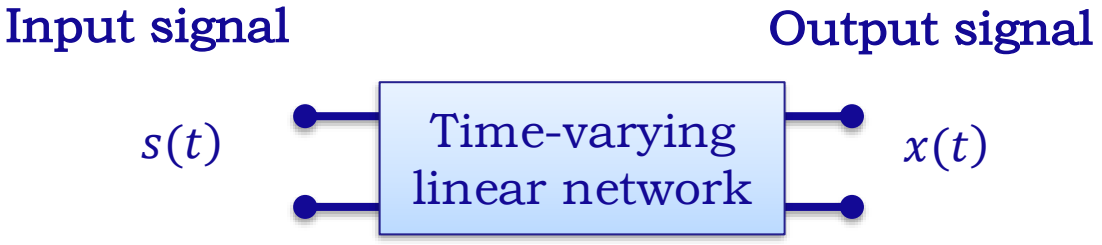
Channel model

- Channel in narrow sense： transmission media
- Channel in wide sense： logical channel
  - Modulation channel
  - Coding channel



Modulation channel model

- Channel characteristics
  - Decay (amplitude)
  - Delay (phase)
- noise interference vs. signal
  - Additive: Independent of signal.
    - Random noise, thermal noise
    - AWGN (additive white Gaussian noise)
  - Multiplicative: dependent of signal
    - Multi-path effect: different amplitude and phase response to different frequency components of the signal, resulting in different attenuation and delay.



$$x(t) = f\left[s(t)\right] + n_i(t)$$
$$= h(t) * s(t) + n_i(t)$$

multiplicative interference

additive interference

$$H(\omega) = \left|H(\omega)\right| e^{-j\omega t_d(\omega)}$$

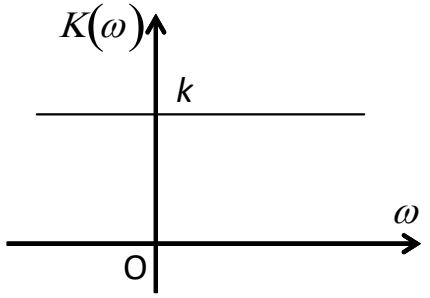
Decay (amplitude) Delay (phase)

Modulation channel model (cont.)

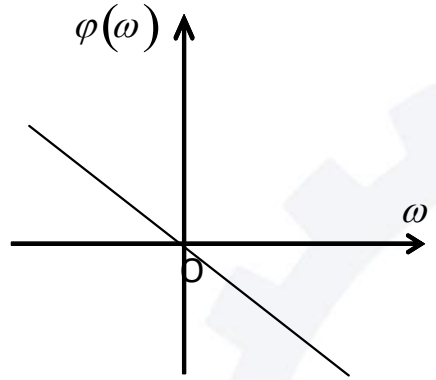
- 信道参量与时间的关系
  - 时不变：恒参信道
    - 理想频率特性：线性时不变系统
    - 实际信道
      - 对称电缆、同轴电缆、光纤
      - 微波中继信道、卫星中继信道
    - 非理性特性
      - equalization
  - 时变：随参信道
    - 传输特性随时间随机变化
    - 实际信道
      - 电力线通信
      - 移动通信、广播、水下声波通信
    - 处理：分集接收，自适应增益控制

$$H(\omega)=|H(\omega)|e^{-j\omega t_d(\omega)}$$

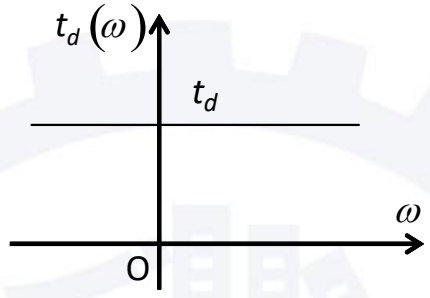
$$|H(\omega)|=k\text{常数} \qquad \varphi(\omega)=-\omega t_d \qquad t_d(\omega)=-\frac{d\varphi(\omega)}{d\omega}=t_d$$



（a）幅度频率特性



（b）相位频率特性



（c）群延迟频率特性

无失真传输条件
- 幅频特性为实常数
- 相频特性为线性

1.4.3 CHANNEL CAPACITY

Channel capacity

- Channel capacity: the maximal error-free transmission rate, C
- Shannon continuous channel capacity

$$C = B\log_2\left(1+\frac{S}{N}\right)=B\log_2\left(1+\frac{S}{n_0 B}\right)$$

- Provided that the actual transfer rate $\boldsymbol{R \le C}$, it is theoretically possible to achieve error-free transmission
- Given $\boldsymbol{C}$, there is a tradeoff between ***SNR*** and **bandwidth (*B*)**
  - A small amount of bandwidth can save signal power obviously
  - It is less effective to save bandwidth increasing the signal power

- Bandwidth B
- Average signal power S
- AWGN channel

  - Average noise power

    $N = n_0 B$

  -Single sideband noise power spectral density n_0

- Note the logarithm function

Channel capacity

$$C = B\log_2\left(1+\frac{S}{N}\right)=B\log_2\left(1+\frac{S}{n_0 B}\right)$$

- To improve the channel capacity
  - Increase the ***SNR*** (either by decreasing ***N*** or increasing ***S***)
    - esp. when $n_0 \to 0$, $C \to \infty$, i.e. unlimited capacity of errorless channel
  - Increase the ***bandwidth (B)***
  - $B \to \infty$, there is a limit of channel capacity

$$\lim_{B\to\infty} C = \frac{S}{n_0}\log_2 e \approx 1.44\frac{S}{n_0}$$

Further discussion goes to
the future *Information Theory* module

CHANNEL AND INTERFERENCE

- Narrow-sense channels only cover communication media.
- The use of wide-sense channel models makes it easy to study modulation or coding problems.
- This course considers additive noise channels.
- Channel capacity is limited, and efficiency and reliability can be compromised.

SUMMARY OF THIS CHAPTER

- Communication is the process of transmitting information, which has the nature of randomness.

  information, entropy
- Communication system modeling analysis is the basic content of this course.

  communication system model: general/analog/digital
- There is a compromise relationship between the reliability and effectiveness of the communication system.

  effective bandwidth, transfer rate, bandwidth efficiency.

  output signal-to-noise ratio, error symbol rate, error bit rate.

Quiz

1. **How to calculate the Channel Capacity?**

2. **What is the difference between “message” and “information”?**

3. **What are the basic modules of a communication system?**

State-of-the-art radio communications

Lancaster University

- Let us watch a video released by IEEE
  http://v.qq.com/page/b/0/x/b03208xqg6x.html
  - A **new record of spectrum efficiency** for 5G
  - Enabled by **massive MIMO** technology
  - **Record high spectrum efficiency @ 145 bits/s/Hz**

Play the video online

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Example applications of modern communication systems

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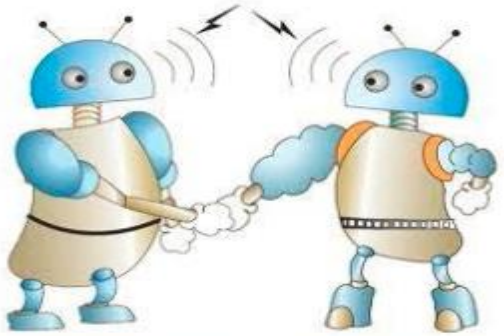
















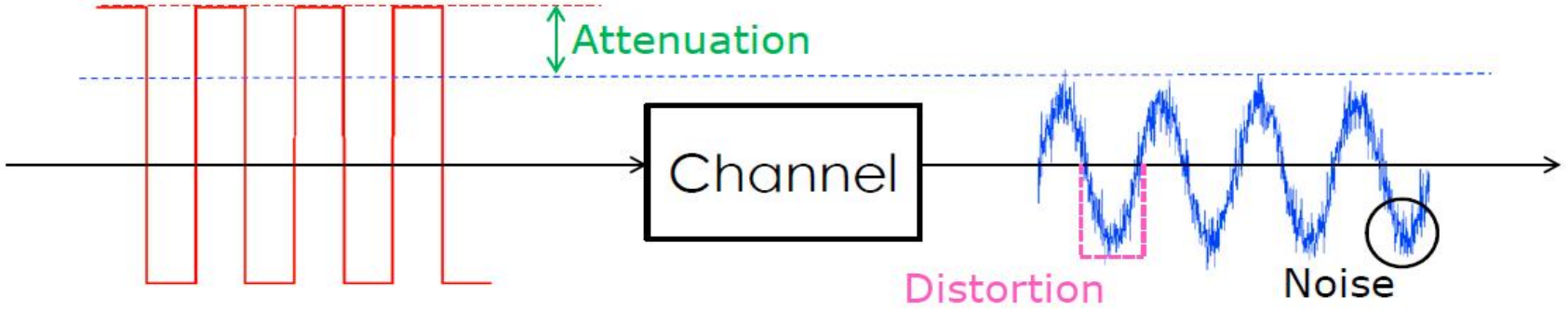
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Communication Channels



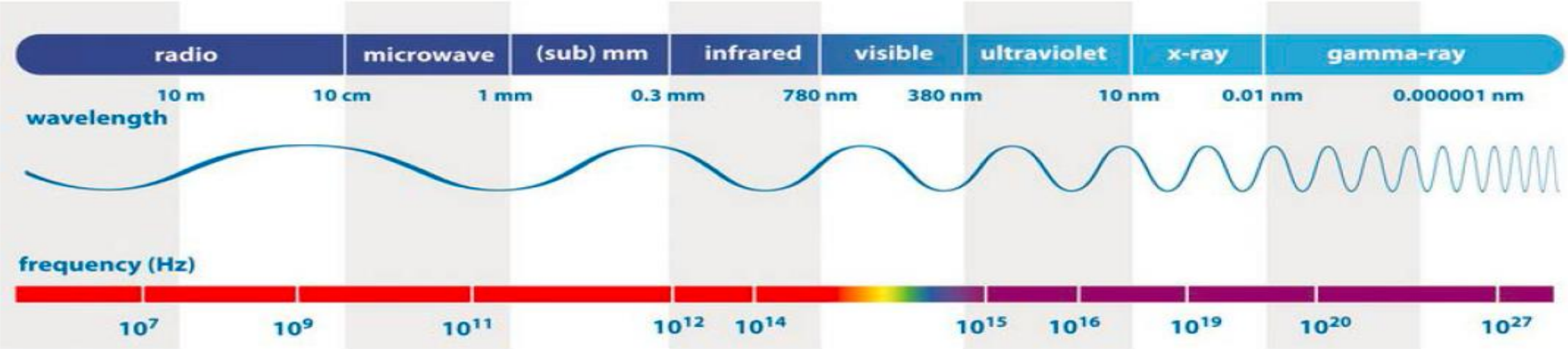



- Common characteristics



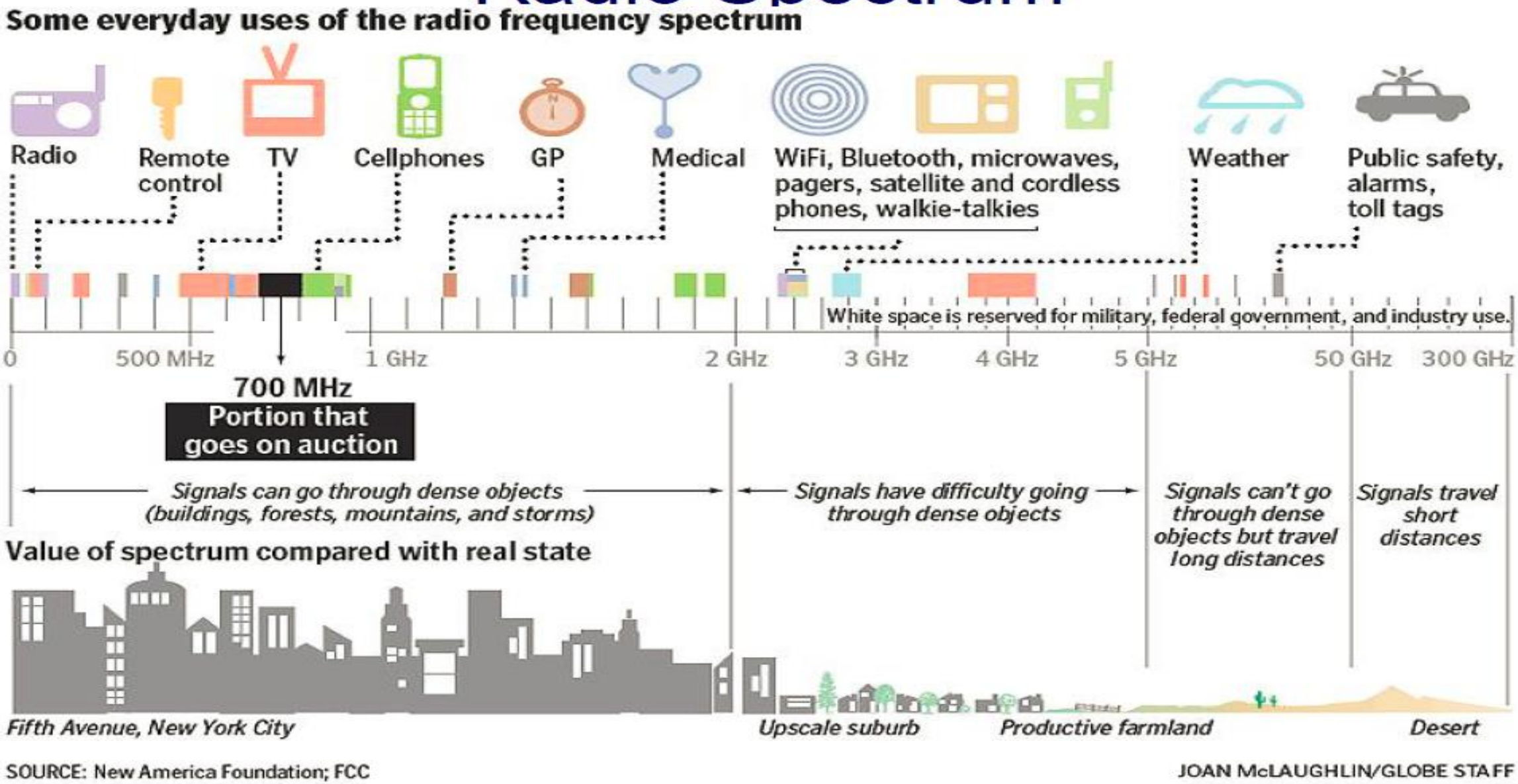
What is the maximum rate at which information can be transmitted across the channel without error?

Radio Spectrum



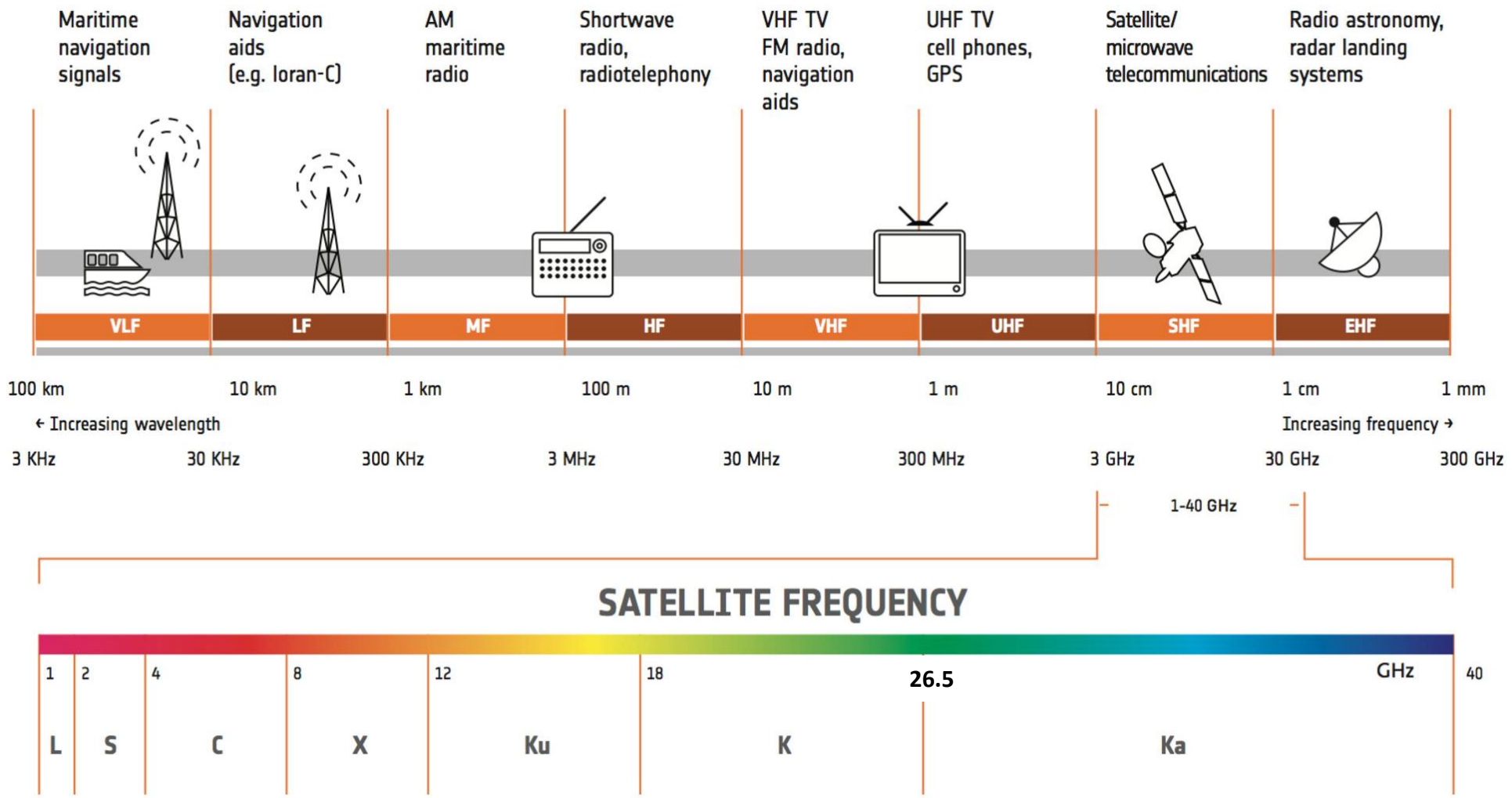
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Radio Spectrum



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Radio Bands (Radio Channels)

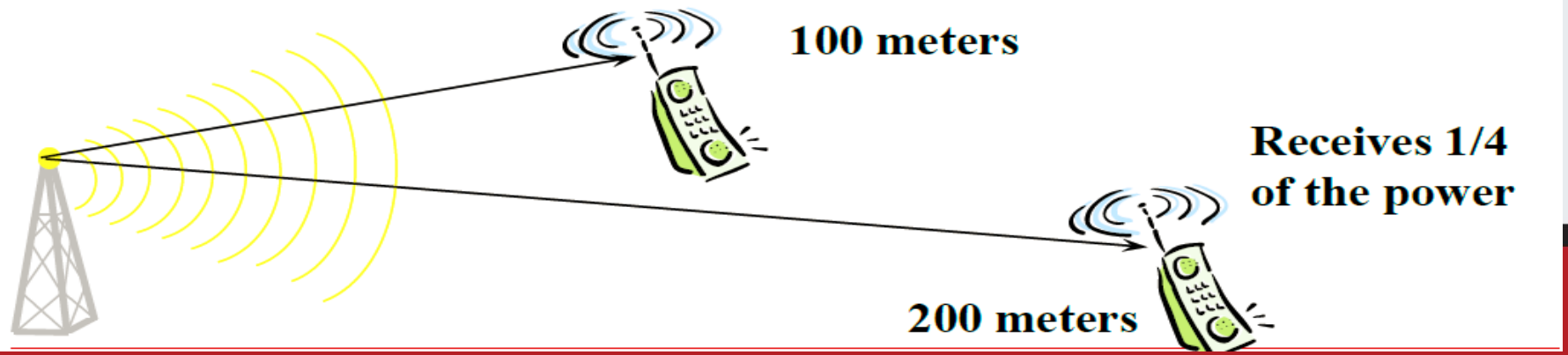


Propagation

- Free-space propagation model

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

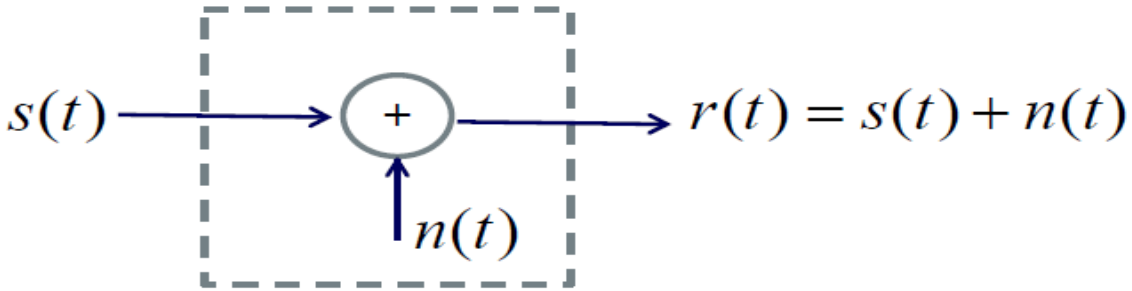
- P_t : transmit power
- P_r : receive power
- G_t: transmit antenna gain
- G_r: receive antenna gain
- λ : wavelength
- d : distance



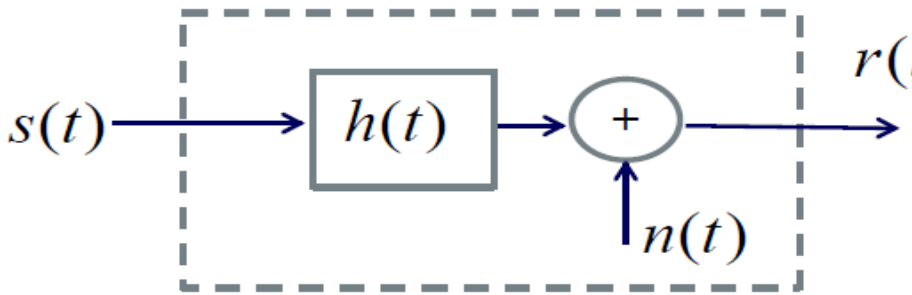
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Mathematical Models

- The additive noise channel



- Linear filter channel

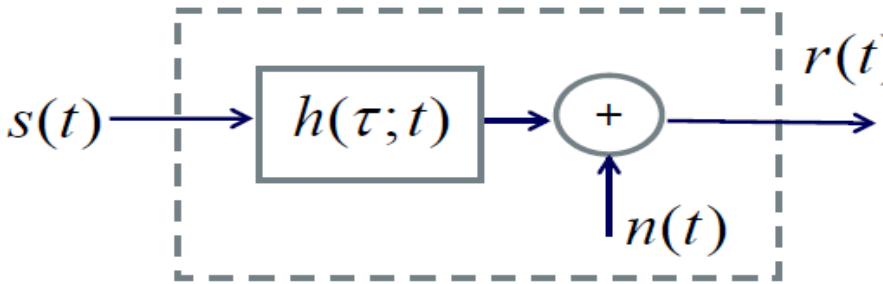


$$r(t) = s(t) * h(t) + n(t)$$

$$= \int_0^{\infty} h(\tau) s(t-\tau) d\tau + n(t)$$

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- Linear time-variant filter channel



$$r(t) = s(t) * h(\tau;t) + n(t)$$

$$= \int_{-\infty}^{\infty} h(\tau;t)s(t-\tau)d\tau + n(t)$$

  - Consider a multi-path signal propagation

$$h(\tau;t) = \sum_{k=1}^{L} a_k(t)\delta(t-\tau_k)$$

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